1 a) Define the **responsivity** and the **Noise Equivalent Power** (NEP) of a radiation detector.

2 **3.** Write the equations that model the capacitance of a MOS capacitor as a function of the gate voltage V_G, measured at low and high frequency, and draw a pictorial plot of the capacitance as a function of V_G. Illustrate the behaviour of the MOS capacitor in the different operating regimes.

3 Illustrate the main properties of a lambertian source.

4 **4.** Photomultiplier tubes and avalanche photodiodes exploit internal amplification to detect extremely low light signals.
(a) Describe the operating principles and the main features of these types of detectors
(b) Critically discuss the main sources of noise in these detectors
(c) Calculate the NEP of a typical photomultiplier, assuming that the dark current is negligible

5 Synthetically describe the main features of thermal detectors. Illustrate the operation principle of pyroelectric detectors.

6 a) **Precisely** define the concepts of Responsivity and Noise Equivalent Power (NEP).
b) Show how to calculate the NEP of a photodetector once the responsivity R and the noise $N_{r.m.s.}$ are known.
c) Sintetically discuss the difference between Thermal Detectors and Quantum Detectors, with pros and co

7 a) Describe the operating regimes of a MOS capacitor used for charge accumulation in CCD and CMOS sensors. Refer each regime to the plot of the MOS capacitance measured at low and high frequency, as a function of the bias voltage V_G.
b) Synthetically illustrate the main noise sources in scientific grade CCD detectors.

8 a) Using a band model, explain the main properties of a Negative Electron Affinity (NEA) photocathode and compare its features with those of a standard semiconductor photocathode;
b) synthetically describe the operating principles of a photomultiplier;
c) illustrate the main noise sources in a photomultiplier;
d) critically compare photomultipliers with photodiodes, in low light condition.

9 Compare avalanche and pin photodiodes, in terms of SNR

10 4) Discuss, using a simple band model the charge accumulation process in a MOS capacitor used as a storage element in a pixel of a CCD/CMOS image sensor. Draw the qualitative plots of the quantities: charge density, electric field and potential as a function of the position in the MOS structure i) at the beginning of charge accumulation process; ii) at the end of the charge accumulation process. Comment the plots.

11 Write the equations that model the capacitance of a MOS capacitor as a function of the gate voltage VG, measured at low and high frequency, and draw a pictorial plot of the capacitance as a function of VG. Illustrate the behaviour of the MOS capacitor in the different operating regimes.

Capacitance CMOS; descrivi i regimi CMOS, con grafici

12 **Omodyne detection**

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